

Use case: an adviser that follows a 5G incident, mapped to Y.3172



The use case

A hospital's private 5G slice: latency (T0) → suspicious logins (T1) → ICU patient data (T2) → “what must we report, to whom, by when?” (T3)

For: operator security and network teams, legal and compliance, regulator liaison

Gives: Indian obligations with deadlines, each quoted and verified; cross-domain links; potential policy gaps

Guardrail: decision support only; a person approves every fix and notice

Example run: attack detected within the deadline • model re-selected once on load drift • 12 draft notices

ML Intent (YAML)

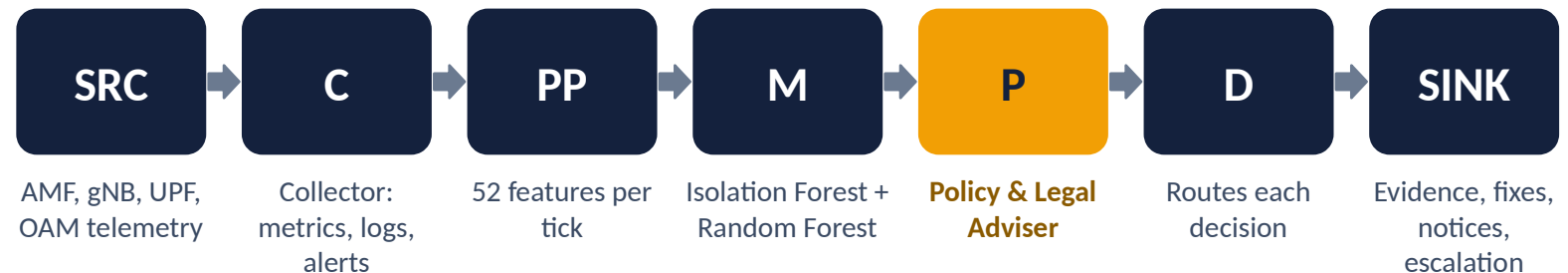
Detect AMF signalling storms:
macro-F1 ≥ 0.85 , p95 ≤ 80 ms,
deadline

MLFO

Instantiates, deploys, monitors, re-selects

ML sandbox

Trains and compares models; tests playbooks before live use



P node = our legal adviser

- 7 specialist agents quote Indian law from their own knowledge bases
- A separate Verifier rates every claim against a Canonical KB
- Blocking mode: no fix runs until a person approves, having seen the obligations

SINKs

- Evidence hashed before any change
- Tiered fixes with human gates
- Draft notices: CERT-In 6 h
- Escalation if unresolved

Knowledge base, mapped to Indian policy and to ITU AI Readiness

23

official sources

8

knowledge bases: 7
agents + Canonical KB

284

documents in the
obligation-check index

12/12

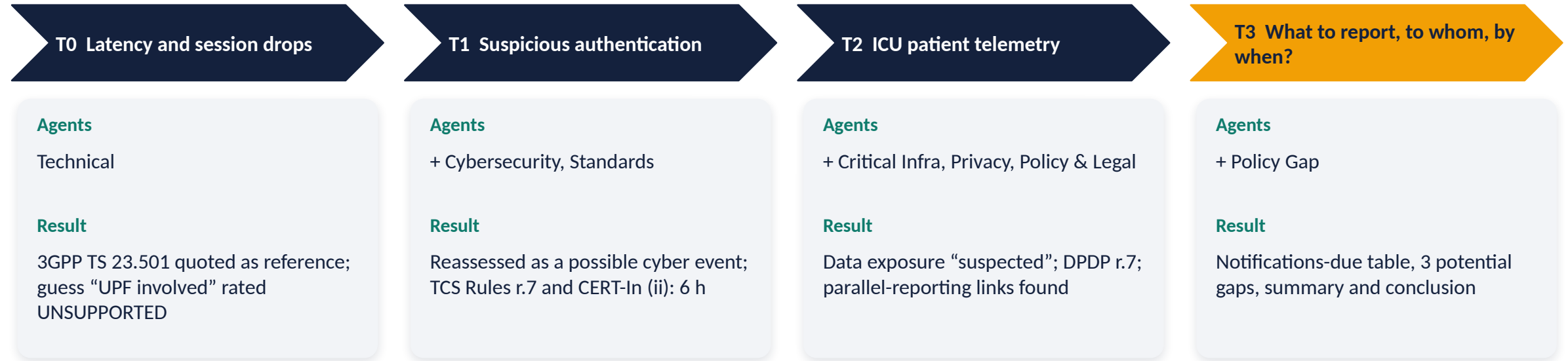
test queries: right
section in top 5

Knowledge base	Key sources	Policy mapping: what it answers	ITU AI Readiness	How the project meets it
Cybersecurity	Telecom Cyber Security Rules 2024 r.7; CERT-In Directions 2022	Report a security incident within 6 h to the Central Government and to CERT-In; keep logs 180 days	Contextualization and regional development	India-specific law and institutions; EU and South Asian comparators labelled by region
Privacy	DPDP Act 2023; DPDP Rules 2025 r.6–7	Breach intimation to the Data Protection Board and affected persons; safeguards	AI integration in domains, cross-domain	7 domain agents; links proven from cited text (e.g. TCS Rules made under Act s.22)
Critical Infrastructure	Telecommunications Act 2023 s.22; CTI Rules 2024	Notified critical telecom infrastructure: 6 h intimation, inspection	Human interaction and supervision	Decision support only; P-node holds, tier gates, review notice on every screen
Policy & Legal	Telecommunications Act 2023; TRAI Act 1997; NDCP 2018	Authorisation, regulator powers, responsible institutions	Policy opportunity identification	Policy Gap agent and a 5-theme gap register, anchors checked word for word
Standards / Technical	3GPP TS 23.501, 33.501; NIST CSF 2.0, SP 800-61; ITU-T Y.3172	Technical reference, labelled “not Indian law”	Open data and open source	Official public texts with URLs; open-source code; free, local stack
Policy Gap	NCSP 2013; NDCP 2018; TRAI AI recommendations; ENISA	Potential gaps, with global and regional comparators		
Canonical KB	All of the above	The Verifier's separate source of truth		

Every passage keeps document, section, page, URL and provenance; a source is refused at build if its text lacks its own title.

Evaluation scenarios, step by step

Scenario 2 — healthcare private 5G slice: the incident is released one stage at a time



Further evaluation scenarios



Results: agent selection matched the official stage table 4/4 • 497 automated tests pass • offline runs replay and re-execute exactly

Most impactful standards gaps, policy gaps and innovations



Standards gaps

Network slicing

Defined in 3GPP TS 23.501 and named by ENISA (slice-as-a-service), yet absent from all 886 Indian legal passages examined

AI-driven network remediation

Y.3172, NIST AI RMF and the EU AI Act frame ML pipelines and oversight; no binding Indian rule found for human oversight of AI fixes



Policy gaps

Four reporting clocks, no single window

CERT-In 6 h, Central Government 6 h, CTI 6 h, DPDP Board without delay; CERT-In and DPDP do not refer to each other

Log retention mismatch

CERT-In: 180 days in India; DPDP Rules: at least 1 year

Who leads?

CTI under Telecom Act s.22 vs NCIIPC: no examined passage addresses both



Innovations

Adviser as the Y.3172 P node

Obligations checked and fixes held for a person before any automated action

Quote-only agents, separate Verifier

Every claim cites document, section, page; rated against a Canonical KB

Per-agent information, staged reassessment

Ends in a “notifications due” answer: who, by when

Auditable by design

Hash-chained trail, replay and re-execution; attacks shown step by step

Gaps are potential gaps for expert review, with Indian anchors verified word for word; never findings of failure.